ಕ

KANNADA DIGITAL INNOVATION HACKATHON 2026

Kannada for Everyone

Learn Kannada. Speak with Confidence.

"Practical Kannada for everyday life, one conversation at a time."
A dedicated, zero-barrier language platform for students, employees, and newcomers in Karnataka.

📍 Organized by HackerRank Campus Crew & NIE Mysuru

🏆 Track: Problem Statement #1 — Kannada for Everyone

SLIDE 2 • PROBLEM STATEMENT

The Real-World Language Barrier in Karnataka

Karnataka, home to Bengaluru — India's premier tech and educational capital — welcomes hundreds of thousands of new students, engineers, and professionals every year.

🚌

Daily Commute Friction

Commuters struggle to hail auto rickshaws, negotiate meters, confirm BMTC bus destinations, and handle change in crowded transit hubs.

☕

Everyday Living & Dining

Ordering food at self-service Darshinis, asking vegetable vendors for rates per kg, and discussing rent with landlords feels intimidating.

💼

Workplace & Social Bonding

Non-Kannada speakers often feel culturally isolated, missing casual cafeteria banter and the warmth of connecting in the local language.

Official Hackathon Objective:
"Develop an interactive platform that helps non-Kannada speakers learn practical Kannada through everyday conversations, situations, vocabulary, and guided practice."

SLIDE 3 • MARKET ANALYSIS

Existing Systems & Educational Gaps

Traditional language tools fail beginner adult learners who need immediate conversational fluency.

❌ Limitations of Current Tools

- **Isolated Vocabulary Lists:** Memorizing words without knowing when or how to say them in a sentence.
- **Academic Grammar Overload:** Complex verb conjugations that intimidate complete beginners.
- **No Real-World Context:** Lack of practical scenarios like auto rickshaws, bus stands, and hospital visits.
- **Sign-up & Paywall Barriers:** Forced account registration before users can learn even a single word.
- **Missing Phonetic Guidance:** Beginners cannot read Kannada script without accurate transliterations.

✅ What Non-Kannadigas Actually Need

- **Dialogue-First Learning:** Hearing how Person A and Person B talk in authentic street settings.
- **Situational Decision Training:** Choosing what to say when an auto driver asks for double meter.
- **Instant Audio Pronunciation:** Web Speech API synthesis with clear Roman transliterations.
- **Zero Friction:** Open the browser and start practicing immediately without logins.
- **Polite Cultural Etiquette:** Knowing when to use "-ri", "-kodi" (respectful) vs "-kodu".

SLIDE 4 • OUR SOLUTION

Our Pedagogical Framework: The 4 Pillars

A structured, beginner-centric learning journey transforming hesitation into conversational confidence.

📖

1. Vocabulary

65+ high-frequency words across 12 categories with flashcards, examples, audio, and bookmarking.

💬

2. Dialogues

10 complete real-world conversations with Person A & B, line-by-line audio, and role-play mode.

🧭

3. Situations

10 real-life survival scenarios (auto, bus stand, Darshini, hospital) with interactive decision challenges.

🏆

4. Practice

Instant feedback quiz engine, streak tracking, unlockable achievement badges, and localStorage persistence.

Key Innovation: Authentic Spoken Kannada + Web Speech API + 100% Offline Persistence

Zero Login Required

SLIDE 5 • IMPLEMENTED MODULES

Core Functional Modules Walkthrough

All 5 core modules are genuinely functional, verified through 14 automated browser tests and live user interactions.

Module 1

Interactive Vocabulary

65+ curated entries across 12 categories. Includes real-time search, 3D flip flashcards, and "Mark as Learned" state.

Module 2

Real-World Dialogues

10 comprehensive everyday conversations (Auto, Darshini, BMTC, Clinic, Office). Line-by-line audio and comprehension check.

Module 3

Scenario Simulations

10 interactive situational scenarios with cultural etiquette tips, partner response expectations, and scenario decision quizzes.

Module 4

Quiz Engine & Review

Categorized tracks (Quick 5-min Sprint, Situational), instant pedagogical explanations, confetti celebration, and question-by-question review.

Module 5

Learner Dashboard

Weighted progress bar, daily practice streak counter, 8 unlockable badges, bookmarked words quick list, and reset modal.

Module 6

Phonetics & Web Speech

Kannada vowels (ಸ್ವರಗಳು), speed controls (0.7x, 0.85x, 1.0x), voice diagnostic check, and transparent educational fallback.

SLIDE 6 • SYSTEM ARCHITECTURE

Technical Stack & Data Architecture

Modern, responsive, lightweight, and engineered for high availability with zero server overhead.

Technology Stack

Frontend Core	**React 18 + Vite 6 (Blazing Fast SPA)**
Design System	**Tailwind CSS v3 + Lucide React**
Typography	**Google Noto Sans Kannada + Inter**
Speech Synthesis	**Browser Web Speech API (kn-IN / en-IN fallback)**
Client Persistence	**HTML5 localStorage (Zero account wall)**
Automated QA	**Puppeteer-core Browser Test Suite (14/14 Pass)**

Component & Data Flow Diagram

```
[ Browser / Client Interface ]
    |
    ├── React State & Router Hook (Tab Navigation)
    |
    ├── Content Layer
    |    ├── vocabulary.js (65+ entries)
    |    ├── conversations.js (10 dialogues)
    |    ├── situations.js (10 interactive challenges)
    |    └── quizzes.js (30 validated items)
    |
    ├── Services Engine
    |    ├── speech.js (Web Speech API kn-IN)
    |    └── storage.js (LocalStorage Persistence)
    |
    └── Gamification (Badges, Streak, Confetti)
```

SLIDE 7 • CONCLUSION & FUTURE SCOPE

Impact, Inclusion & The Road Ahead

"Kannada for Everyone" removes language anxiety, celebrates Karnataka's culture, and empowers newcomers to feel truly at home.

Demonstrated Impact

- **Zero Friction Onboarding:** Anyone can start learning in 3 seconds directly on their smartphone or laptop.
- **Street-Smart Usability:** Prepares learners for actual Bengaluru situations: bargaining, hiring autos, ordering dosas, and emergency calls.
- **Promotes Harmony:** Fosters mutual appreciation between native Kannadigas and non-Kannada speaking residents.

Future Roadmap

- **Regional Dialect Explorer:** Add North Karnataka (Hubballi-Dharwad) and Coastal Kannada (Kundapura/Mangaluru) variations.
- **Speech Recognition Pronunciation Scoring:** Enable users to speak into the microphone and receive instant pronunciation feedback.
- **Offline PWA Support:** Progressive Web App installable on Android and iOS devices with full offline caching.

ಧನ್ಯವಾದಗಳು! (Thank You!)

Code. Create. Celebrate Kannada. • NIE Mysuru & HackerRank Campus Crew